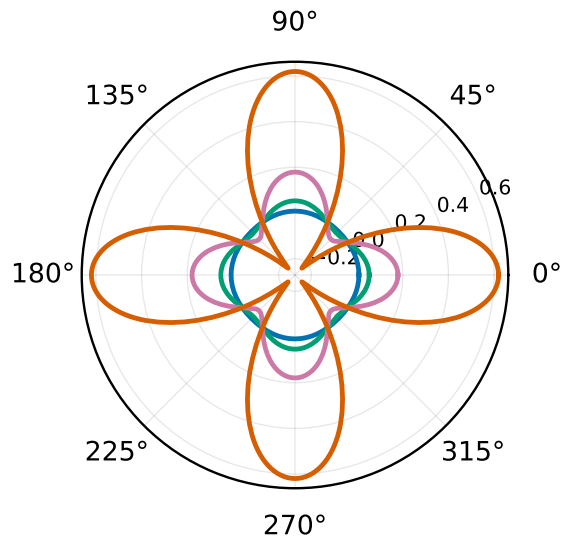
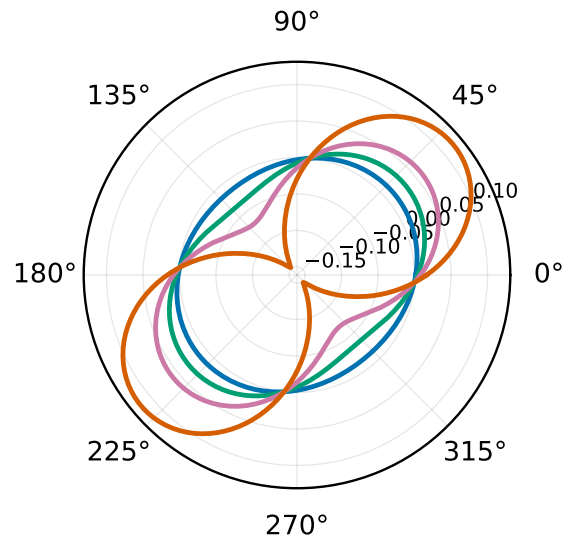




K4 photon (quartic): $\omega/(c|k|) - 1$ vs direction



random control (quadratic): $\omega/(c|k|) - 1$ vs direction

— $kl = 1.00$
— $kl = 1.60$
— $kl = 2.20$
— $kl = 3.14$
— $kl = 1.00$
— $kl = 1.60$
— $kl = 2.20$
— $kl = 3.14$